

LightningParse: Hybrid Rust-Python PDF Extraction

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Abstract

We present LightningParse, combining Rust parallel extraction with Python NLP integration. Achieves 47x median speedup on digital-native PDFs.

I. Introduction

Recent advances in document parsing have demonstrated that parallel extraction architectures can significantly outperform traditional sequential approaches.

II. Architecture

The proposed pipeline operates in two tiers. Tier 1 handles digitally-authored PDFs by directly parsing the content stream operators.

III. Extraction

Experimental evaluation on a corpus of 50 documents shows a median speedup of 47x over baseline Python libraries.

IV. OCR Fallback

Font metric resolution is critical for accurate bounding box computation.

V. Font Metrics

Header and footer detection uses a heuristic approach based on repeated text in page margins.

VI. Headers

Reading order reconstruction sorts blocks by vertical position then applies column-detection.

VII. Reading Order

The FFI boundary between Rust and Python uses JSON serialization with less than 2 percent overhead.

VIII. Results

Future work includes structured table extraction for grid-aligned text blocks.

IX. Discussion

Recent advances in document parsing have demonstrated that parallel extraction architectures can significantly outperform traditional sequential approaches.

X. Related Work

The proposed pipeline operates in two tiers. Tier 1 handles digitally-authored PDFs by directly parsing the content stream operators.

XI. Conclusion

Experimental evaluation on a corpus of 50 documents shows a median speedup of 47x over baseline Python libraries.

XII. Future Work

Font metric resolution is critical for accurate bounding box computation.

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